

Computer Networks Exam Solution Key

I. SECTION 1: Multiple Choice Questions (MCQs 1-70)

NAT & PAT (Questions 1-15)

1. Which of the following ranges is defined as a non-routable private IP address space?
 - A. 11.0.0.0 – 11.255.255.255
 - B. 172.16.0.0 – 172.31.255.255
 - C. 192.167.0.0 – 192.167.255.255
 - D. 224.0.0.0 – 239.255.255.255

Correct Answer: B

2. What is the technically accurate term for NAT Overload?
 - A. Static Network Address Translation
 - B. Dynamic Host Configuration Protocol
 - C. Port Address Translation (PAT)
 - D. Round-Robin IP Distribution

Correct Answer: C

3. In the context of the NAT Translation Table, “Inside Local” refers to:
 - A. The public IP address assigned to the router’s WAN interface.
 - B. The actual private IP address assigned to a host within the internal network.
 - C. The IP address of the destination server on the public internet.
 - D. The MAC address of the internal gateway.

Correct Answer: B

4. Why is Port Address Translation (PAT) considered the most cost-effective NAT implementation?
 - A. It eliminates the need for Layer 3 routing.
 - B. It allows thousands of internal hosts to share a single registered public IP address.
 - C. It requires no processing overhead for checksum recalculations.
 - D. It automatically encrypts all payload data.

Correct Answer: B

5. Which NAT technique creates a permanent, one-to-one mapping between a local IP and a global IP?
- A. Dynamic NAT
 - B. Static NAT
 - C. IP Masquerading
 - D. PAT

Correct Answer: B

6. What mechanism does a NAT router use to maintain the mapping between internal requests and external responses?
- A. Routing Information Base (RIB)
 - B. NAT Translation Table
 - C. ARP Cache
 - D. MAC Address Table

Correct Answer: B

7. The private IP range 10.0.0.0 through 10.255.255.255 is colloquially known as:
- A. The "10-dot" range
 - B. The Loopback range
 - C. The APIPA range
 - D. The Multicast range

Correct Answer: A

8. When a NAT device modifies an IP address in the packet header, it must also update the:
- A. Ethernet Preamble
 - B. IP header checksum
 - C. Time to Live (TTL) value
 - D. Frame Check Sequence (FCS)

Correct Answer: B

9. How does PAT resolve a conflict when two internal hosts use the same source port to access the same destination?
- A. It drops the segment from the second host.
 - B. It assigns a unique public-side source port number to each session.
 - C. It requires the internal hosts to re-negotiate their MAC addresses.
 - D. It buffers the second packet until the first session terminates.

Correct Answer: B

10. The classification of NAT mapping that allows "Many-to-1" communication is:

- A. Static NAT
- B. Round-Robin Mapping
- C. PAT / NAT Overload
- D. Unicast Mapping

Correct Answer: C

11. A significant limitation of NAT regarding Peer-to-Peer (P2P) networking is:

- A. Internal hosts cannot use port numbers above 1024.
- B. External hosts generally cannot initiate a connection to a hidden internal host.
- C. NAT routers cannot process UDP traffic.
- D. P2P traffic bypasses the NAT Translation Table.

Correct Answer: B

12. In a Linux environment, which package is utilized to configure NAT and packet filtering rules?

- A. Apache2
- B. OpenSSH
- C. Netfilter/iptables
- D. BIND9

Correct Answer: C

13. Which iptables target is specifically used to implement IP Masquerading for dynamic interfaces?

- A. -j ACCEPT
- B. -j SNAT
- C. -j MASQUERADE
- D. -j DROP

Correct Answer: C

14. The "Inside Global" address represents:

- A. The private address used by a host to communicate within the LAN.
- B. The public IP address as it appears to the outside world.
- C. The internal IP address of the gateway router.
- D. The hardware address of the external web server.

Correct Answer: B

15. Which technique distributes incoming connections from a single global IP to a pool of local servers sequentially?

- A. IP Masquerading
- B. Static 1-to-1 Mapping

- C. Round-Robin Mapping
- D. NATP

Correct Answer: C

Transport Layer (Questions 16-30)

16. The Transport Layer provides logical communication between:

- A. Physical network links
- B. Application processes
- C. Intermediate routers
- D. Network interface cards

Correct Answer: B

17. In the "household analogy" for networking, the "kids" sending letters represent:

- A. Hosts
- B. Processes
- C. Sockets
- D. Routers

Correct Answer: B

18. Which transport protocol is characterized by "no-frills," unreliable, and unordered delivery?

- A. TCP
- B. HTTP
- C. UDP
- D. FTP

Correct Answer: C

19. The interface between the application layer and the transport layer is the:

- A. Port
- B. Socket
- C. Segment
- D. Datagram

Correct Answer: B

20. Which layer is responsible for breaking application messages into segments at the sender and reassembling them at the receiver?

- A. Network Layer

- B. Transport Layer
- C. Data Link Layer
- D. Session Layer

Correct Answer: B

21. TCP provides all the following services except:

- A. Flow control
- B. Congestion control
- C. Throughput guarantees
- D. Connection-oriented setup

Correct Answer: C

22. In the household analogy, Ann and Bill (the individuals distributing mail to the kids) represent:

- A. Network Protocols
- B. Transport Protocols
- C. Physical Layers
- D. IP Addresses

Correct Answer: B

23. What is the primary action taken by the sender-side transport layer?

- A. Routing segments to the correct ISP.
- B. Encapsulating application messages into segments.
- C. Determining the MAC address of the next hop.
- D. Framing bits for physical transmission.

Correct Answer: B

24. Multiplexing and demultiplexing of data are primary responsibilities of the:

- A. Physical Layer
- B. Transport Layer
- C. Network Layer
- D. Link Layer

Correct Answer: B

25. UDP is best described as a “no-frills” extension of which “best-effort” protocol?

- A. HTTP
- B. IP
- C. TLS
- D. ICMP

Correct Answer: B

26. When two processes on the same host communicate, they use:

- A. TCP/UDP sockets
- B. Inter-process communication (IPC)
- C. The default gateway
- D. Network Address Translation

Correct Answer: B

27. Which type of application architecture involves processes that act as both client and server?

- A. Client-Server
- B. Peer-to-Peer (P2P)
- C. Master-Slave
- D. Centralized Mainframe

Correct Answer: B

28. In the network stack, the application developer typically has control over:

- A. The internal mechanics of the Transport Layer.
- B. The Socket and the Application Layer.
- C. The IP routing table.
- D. The Network Interface Card (NIC) driver.

Correct Answer: B

29. What is the protocol data unit (PDU) of the transport layer?

- A. Frame
- B. Packet
- C. Segment
- D. Bit

Correct Answer: C

30. Which protocol ensures data integrity and manages end-to-end congestion control?

- A. UDP
- B. ARP
- C. TCP
- D. RARP

Correct Answer: C

Web and HTTP Protocols (Questions 31-45)

31. HTTP is described as “stateless” because:

- A. It does not allow for persistent connections.
- B. The server maintains no information about past client requests.
- C. It cannot be used for dynamic content.
- D. It requires an encrypted session for every request.

Correct Answer: B

32. In non-persistent HTTP, how many RTTs are required to download a single object (excluding transmission time)?

- A. 1 RTT
- B. 2 RTTs
- C. 3 RTTs
- D. 4 RTTs

Correct Answer: B

33. Which version of HTTP introduced persistent connections as the default to reduce latency?

- A. HTTP/1.0
- B. HTTP/1.1
- C. HTTP/0.9
- D. HTTP/3

Correct Answer: B

34. The purpose of the HTTP HEAD method is to:

- A. Upload a new file to the server.
- B. Request only the header portion of a resource.
- C. Delete a resource from the server.
- D. Establish a persistent tunnel.

Correct Answer: B

35. The HTTP status code “404 Not Found” belongs to which category?

- A. Informational
- B. Success
- C. Client Error
- D. Server Error

Correct Answer: C

36. Head-of-Line (HOL) blocking in HTTP/1.1 occurs when:

- A. The server's CPU is overloaded.
- B. A large object delays the transmission of smaller objects behind it in the queue.
- C. The network interface card fails.
- D. The DNS resolution takes too long.

Correct Answer: B

37. HTTP/2 mitigates HOL blocking at the application layer by using:

- A. Non-persistent connections.
- B. Binary framing and interleaving of objects.
- C. Multiple physical links.
- D. Compression of IP headers.

Correct Answer: B

38. HTTP/3 improves performance by moving from TCP to which transport protocol?

- A. SCTP
- B. UDP (via QUIC)
- C. ICMP
- D. RSVP

Correct Answer: B

39. What is the primary objective of a Web Cache (Proxy Server)?

- A. To encrypt user passwords.
- B. To satisfy client requests locally without involving the origin server.
- C. To assign public IP addresses to internal hosts.
- D. To scan all traffic for physical layer errors.

Correct Answer: B

40. When using a Conditional GET, what status code does the server return if the object has not changed?

- A. 200 OK
- B. 301 Moved Permanently
- C. 304 Not Modified
- D. 403 Forbidden

Correct Answer: C

41. Persistent HTTP is analogous to:

- A. Driving to a store for every single item on a shopping list individually.
- B. Driving to the store once and buying all items before returning home.
- C. Hiring a delivery service for each item.

D. Calling the store to check if items are in stock.

Correct Answer: B

42. Round Trip Time (RTT) is defined as:

- A. The time it takes for a packet to travel around a local ring topology.
- B. The time required for a small packet to travel from client to server and back.
- C. The time a server remains idle between requests.
- D. The total time to transmit a 100KB file.

Correct Answer: B

43. Which HTTP method is used to upload user-generated data specifically within the message body?

- A. GET
- B. POST
- C. PUT
- D. OPTIONS

Correct Answer: B

44. Web caches use the "If-modified-since:" header to implement:

- A. Persistent connections.
- B. Conditional GET.
- C. Cookie authorization.
- D. Server-side pushing.

Correct Answer: B

45. Which port is conventionally reserved for unencrypted HTTP traffic?

- A. 21
- B. 443
- C. 80
- D. 53

Correct Answer: C

Network Security & Firewalls (Questions 46-60)

46. When a firewall blocks a packet and sends an ICMP error message to the sender, it is performing a:

- A. Drop

- B. Reject
- C. Accept
- D. Bypass

Correct Answer: B

47. The silent blocking of traffic by a firewall, providing no feedback to the source, is called:

- A. Reject
- B. Drop
- C. Quarantine
- D. Filter

Correct Answer: B

48. Which firewall type tracks the context of active sessions to make filtering decisions?

- A. Static Packet Filter
- B. Stateful Inspection Firewall
- C. Circuit-Level Gateway
- D. Host-Based Firewall

Correct Answer: B

49. Stateful inspection technology was pioneered in the early 1990s by:

- A. Jeff Mogul
- B. Gil Shwed
- C. Marcus Ranum
- D. Paul Vixie

Correct Answer: B

50. Packet-filtering technology was developed at Digital Equipment Corp (DEC) in the:

- A. Late 1970s
- B. Late 1980s
- C. Mid 1990s
- D. Early 2000s

Correct Answer: B

51. A Web Application Firewall (WAF) is specifically designed to protect against:

- A. Layer 3 IP spoofing.
- B. SQL injection and cross-site scripting (XSS).
- C. Physical wiretapping.
- D. MAC address duplication.

Correct Answer: B

52. What is the security best practice for a firewall's "Default Policy"?

- A. Accept all traffic.
- B. Drop or Reject all traffic not explicitly permitted.
- C. Log all traffic but allow it.
- D. Encrypt all traffic by default.

Correct Answer: B

53. Next-Generation Firewalls (NGFW) differ from traditional firewalls by adding:

- A. Faster port speeds.
- B. Deep-packet inspection and application-level awareness.
- C. Support for fiber-optic cables.
- D. Wireless access point functionality.

Correct Answer: B

54. A Proxy Firewall acts as a gateway at which OSI layer?

- A. Layer 2 (Link)
- B. Layer 3 (Network)
- C. Layer 7 (Application)
- D. Layer 1 (Physical)

Correct Answer: C

55. Where is a host-based firewall located?

- A. On the perimeter gateway router.
- B. On an individual computer or end-system.
- C. At the Internet Service Provider's core.
- D. Inside the Physical Layer hardware.

Correct Answer: B

56. A Circuit-Level Gateway validates security by vetting:

- A. Individual data payloads.
- B. Ongoing TCP/UDP connections (sessions).
- C. Hardware serial numbers.
- D. Electrical signal strength.

Correct Answer: B

57. Who is credited with developing the graphical interface for Firewall-1, making the technology accessible?

- A. Brian Reid
- B. Nir Zuk
- C. Marcus Ranum
- D. Paul Vixie

Correct Answer: B

58. Unified Threat Management (UTM) systems typically integrate:

- A. Router, Switch, and Bridge.
- B. Firewall, IDS/IPS, and Antivirus.
- C. Monitor, CPU, and Keyboard.
- D. HTTP, FTP, and SMTP.

Correct Answer: B

59. Standard Packet Filtering firewalls primarily operate at which layer?

- A. Layer 2
- B. Layer 3
- C. Layer 5
- D. Layer 7

Correct Answer: B

60. The primary disadvantage of deep-packet inspection (DPI) in firewalls is:

- A. It is less secure than packet filtering.
- B. It can significantly impact network performance and latency.
- C. It cannot read IP addresses.
- D. It only supports the Linux operating system.

Correct Answer: B

Web Utilities & IP Services (Questions 61-70)

61. Which of the following is an essential component of the HTTP cookie system?

- A. Physical Layer Security
- B. Back-end database at the website
- C. Recursive DNS server
- D. NAT Translation Table

Correct Answer: B

62. RARP (Reverse Address Resolution Protocol) was limited because:

- A. It was too fast for Ethernet.
- B. It only assigned the IP address, not the gateway or subnet mask.
- C. It only worked on wireless networks.
- D. It required manual port assignment for every packet.

Correct Answer: B

63. In what year was the BOOTP protocol introduced?

- A. 1975
- B. 1985
- C. 1995
- D. 2005

Correct Answer: B

64. Which UDP ports are used by BOOTP and its successor, DHCP?

- A. 20 and 21
- B. 53 and 67
- C. 67 (server) and 68 (host)
- D. 80 and 443

Correct Answer: C

65. Dynamic IP assignment is desirable primarily because it:

- A. Makes IP addresses permanent.
- B. Supports mobility and on-demand allocation of limited addresses.
- C. Increases the total number of IPv4 addresses.
- D. Prevents the use of NAT.

Correct Answer: B

66. Transport Layer Security (TLS) provides which services for TCP?

- A. Physical layer shielding.
- B. Encryption, data integrity, and authentication.
- C. Automatic IP address assignment.
- D. Routing path optimization.

Correct Answer: B

67. Where is TLS typically implemented in the networking model?

- A. Network Layer
- B. Application Layer (via specialized libraries)
- C. Data Link Layer
- D. Physical Layer

Correct Answer: B

68. Why are third-party tracking cookies considered a privacy risk?

- A. They can format the user's hard drive.
- B. They allow a common identity to be tracked across multiple different websites.
- C. They increase the power consumption of the computer.
- D. They bypass the host-based firewall.

Correct Answer: B

69. Which application-layer protocol provides reliability for HTTP/3 over UDP?

- A. TCP
- B. QUIC
- C. RARP
- D. BOOTP

Correct Answer: B

70. What is the approximate limit of the global IPv4 address space?

- A. 1.5 billion
- B. 4.3 billion
- C. 128 billion
- D. Unlimited

Correct Answer: B

II. SECTION 2: Descriptive Questions (Questions 1-9)

1. HTTP Response Time Calculation

Question: A webpage consists of one base HTML file and multiple referenced images. Let N represent the total number of objects (HTML + images). Formulate and explain the total response time required to download this page using non-persistent HTTP.

Detailed Solution: In non-persistent HTTP, each object requires a dedicated TCP connection. The process for each object is as follows:

- **Step 1: TCP Handshake:** The client initiates a connection. This requires one Round Trip Time (RTT).
- **Step 2: HTTP Request/Response:** The client requests the object, and the server transmits it. This requires a second RTT plus the time needed to transmit the file (transmission time).

Total Per Object: Each individual object takes $(2 \text{ RTT} + \text{transmission time})$.

Final Formula: For a webpage with N total objects: $\text{Total Time} = N \times (2 \text{ RTT} + \text{transmission time})$.

Note: If the question defines N specifically as the number of images (and 1 HTML file), the formula would be $(1 + N) \times (2 \text{ RTT} + \text{transmission time})$.

2. Web Caching and Link Utilization Performance

Question: An institution has a 1.54 Mbps access link. The request rate results in an average data rate of 1.50 Mbps. Calculate the current link utilization and evaluate the performance of installing a web cache with a 40% hit rate compared to a link upgrade.

Detailed Solution:

- **Current Link Utilization:** $1.50 \text{ Mbps} / 1.54 \text{ Mbps} = 97.4\%$. At this level, the link suffers from massive queuing delays.
- **Implementing a Web Cache (40% hit rate):**
 - **Traffic Reduction:** 40% of requests are satisfied locally by the cache. Only 60% of traffic must traverse the access link.
 - **Adjusted Data Rate:** $0.60 \times 1.50 \text{ Mbps} = 0.9 \text{ Mbps}$.
 - **New Access Link Utilization:** $0.9 / 1.54 = 58.4\%$.
- **Performance Analysis:** By reducing utilization to 58%, queuing delays are minimized. The resulting average end-to-end delay is approximately 1.2 seconds. This 1.2s figure accounts for the near-zero delay of the 40% cache hits combined with the RTT and transmission time for the 60% of requests that must still reach the origin server. This is a significantly more cost-effective solution than upgrading to a 154 Mbps link.

3. The Mechanics of Port Address Translation (PAT)

Question: Detail the mechanism a PAT-enabled router employs to maintain connection uniqueness when multiple internal hosts use identical source port numbers.

Detailed Solution:

- **Address Translation:** The router manages the transition between the "Inside Local" (private IP) and the "Inside Global" (public IP) addresses.
- **Port Modification:** If Host A (10.0.1.2:1000) and Host B (10.0.1.3:1000) both send requests, the router cannot simply map both to its public IP on port 1000, as returning traffic would be ambiguous.

- **The Translation Process:**

- (a) The router replaces the private source IP with its own public source IP.
- (b) The router replaces the private source port with a unique public source port (e.g., mapping Host A to 2001 and Host B to 2002).

- **NAT Translation Table:** The router stores these mappings (e.g., 10.0.1.2 : 1000 ↔ PublicIP : 2001).

- **Return Path:** When the server replies to PublicIP:2001, the router consults the table, identifies Host A as the recipient, and restores the original private IP and port.

4. Comparative Analysis: Network Layer vs. Application Layer Inspection

Question: Compare firewall inspection at Layer 3 (Network) versus Layer 7 (Application).

Detailed Solution:

Feature	Network Layer (Layer 3)	Application Layer (Layer 7)
Depth of Inspection	Basic: Examines IP addresses, protocols (TCP/UDP), and ports.	Deep: Inspects actual payload content (HTTP requests, emails).
Speed vs. Security	High Speed: Low processing overhead but limited security.	Comprehensive Security: Slower due to deep-packet inspection (DPI).
Threat Detection	Identifies IP Spoofing and unauthorized port access.	Detects SQL Injection, XSS, and malware hidden in payloads.
PDU Handled	IP Datagrams/Packets	Application Messages / Data Payloads

5. Transport Layer: The Household Analogy

Question: Map the components of the “household analogy” to their respective network layer and transport layer counterparts.

Detailed Solution: The analogy effectively illustrates how the transport layer enhances the service of the network layer:

- **Hosts (Houses):** The end-systems where communication terminates.
- **Processes (Kids):** The specific application instances requiring communication.
- **App Messages (Letters in Envelopes):** The raw data generated by the applications.
- **Transport Protocol (Ann and Bill):** The entity that demultiplexes messages to specific processes (kids) within the host (house).
- **Network Protocol (Postal Service):** The mechanism that provides logical communication between hosts (moving mail between houses).

6. Evolution of HTTP: Solving Head-of-Line (HOL) Blocking

Question: Explain how the evolution from HTTP/1.1 to HTTP/3 addressed the problem of Head-of-Line (HOL) blocking at different layers.

Detailed Solution:

- **HTTP/1.1:** Suffers from Application-layer HOL blocking. Because it uses First-Come-First-Served scheduling on a single connection, a large object (like a high-res image) must finish transmitting before smaller objects behind it can be sent.
- **HTTP/2:** Solves application-layer HOL blocking by introducing Binary Framing. Objects are divided into independent frames and interleaved. This allows multiple objects to be multiplexed over a single TCP connection simultaneously.
- **HTTP/3:** Addresses Transport-layer HOL blocking. In HTTP/2, if a single TCP packet is lost, the entire connection stalls until it is retransmitted. HTTP/3 moves to UDP (using QUIC), which provides per-object error control. A loss in one stream does not stall other independent streams, eliminating the TCP-level stalling issue.

7. Maintaining State: The Four Components of Cookies

Question: Detail the four essential components of the cookie system that allow the stateless HTTP protocol to maintain user session state.

Detailed Solution:

- (a) **Response Header Line (Set-Cookie):** When a user first visits, the server generates a unique ID and sends it to the browser via the `set-cookie:` header in the HTTP response.
- (b) **Request Header Line (Cookie):** For all subsequent requests to that same server, the browser automatically includes the assigned ID in the `cookie:` header.
- (c) **Local Cookie File:** The browser stores the cookie ID on the user's host machine, managed by the client application.
- (d) **Back-end Database:** The website maintains a database linking the unique cookie ID to user preferences, shopping carts, or login sessions.

8. Dynamic IP Assignment: RARP vs. BOOTP vs. DHCP

Question: Compare the historical and functional differences between RARP, BOOTP, and DHCP.

Detailed Solution:

- **RARP (Reverse ARP):** An early protocol where a host broadcasts its MAC address to request an IP. Its major limitation was that it only provided the IP address, failing to provide the subnet mask or default gateway.
- **BOOTP (Bootstrap Protocol):** Introduced in 1985 to improve upon RARP. It operates over UDP (ports 67/68) and provides a more complete configuration, including the IP address, subnet mask, and default router, at boot time.
- **DHCP:** The modern standard and successor to BOOTP. It introduces dynamic on-demand allocation with temporary leases. This supports device mobility (e.g., laptops moving between networks) and efficient IP address reuse, which BOOTP's static assignment lacked.

9. Network Security: The History and Purpose of Firewalls

Question: Summarize the historical evolution of firewalls and the primary actions they take to enforce security policy.

Detailed Solution:

- **Evolutionary Timeline:**

- **Late 1980s:** Jeff Mogul, Brian Reid, and Paul Vixie (DEC) developed the first packet-filtering technology.
- **1991-1992:** Marcus Ranum introduced security proxies (Application-layer firewalls).
- **1993-1994:** Gil Shwed (Check Point) pioneered stateful inspection, while Nir Zuk developed the graphical interface for Firewall-1 to make these systems manageable for businesses.
- **Core Firewall Actions:**
 - **Accept:** Allows the packet to pass through the gateway.
 - **Reject:** Blocks the packet and proactively informs the sender with an error message.
 - **Drop:** Silently discards the packet, providing no feedback to the source.
- **Best Practice:** The “Default Policy” should always be set to Drop or Reject to ensure that any traffic not explicitly permitted by a security rule is denied.